



A Hybrid Modelling Architecture for Predicting Rheological Performance of Waste Frying Oil–Modified Asphalt Binders via Stochastic–Physical Integration

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Abstract: The intrinsic variability associated with waste-derived bitumen modifiers poses persistent limitations for conventional deterministic pavement design approaches. This study establishes a hybrid modelling architecture that integrates physics-informed boundary constraints with stochastic simulation components to predict the rheological performance of asphalt binders modified with waste frying oil. Literature-derived parameter ranges are embedded within a Gaussian copula-based dependency structure and expanded through large-scale Monte Carlo simulation to generate a statistically convergent dataset comprising 1,000,000 realizations. The integrated modelling structure preserves the nonlinear relationships among key rheological indicators, including penetration, softening point, and modifier dosage, while maintaining physical consistency. The predictive capability of the proposed architecture is validated using standard statistical metrics, achieving coefficients of determination exceeding 0.98 for penetration and 0.93 for softening point. A mode-based master curve is further constructed to provide a stable and computationally efficient representation of binder behaviour across varying modifier contents. The results demonstrate that the proposed hybrid modelling structure offers a reliable alternative to extensive laboratory testing, reducing experimental effort while retaining predictive fidelity. From a system perspective, the framework provides a structured basis for incorporating material variability into pavement performance evaluation and can be interpreted as a simplified digital twin component for asphalt material systems. The study thereby contributes to the integration of stochastic modelling within engineering-oriented predictive structures for sustainable pavement design.

Keywords: Hybrid modelling architecture; Stochastic–physical integration; Monte Carlo simulation; Gaussian copula; Bitumen rheology; Digital twin systems

1 Introduction

The sustainable management of hazardous industrial by-products remains a critical challenge in contemporary infrastructure engineering. Among these materials, waste frying oil has attracted increasing attention due to its high acidity, oxidative instability, and the environmental risks associated with improper disposal. From a pavement engineering perspective, however, waste frying oil contains chemical components comparable to the maltene fraction of bitumen, making it a viable candidate for binder modification and rejuvenation [1]. Experimental investigations, particularly those reported by Eriskin et al. [2], have demonstrated that the incorporation of waste oils can significantly influence the rheological properties of asphalt binders, improving low-temperature performance while reducing the required optimum bitumen content. Subsequent studies have further explored modification strategies based on waste-derived additives and rejuvenators. Uyar and Akbulut [3] examined boric-acid-modified bitumen using dynamic shear rheometer and Fourier transform infrared spectroscopy techniques, highlighting the continued importance of experimentally grounded binder characterization. Similarly, Eltwati et al. [4] evaluated the effects of waste engine oil combined with crumb rubber rejuvenators on reclaimed asphalt pavement binders, confirming the functional relevance of waste-derived oils in sustainable material design. In addition, the use of hydrogel microcapsules containing industrial waste oil has been proposed as an effective strategy for enhancing binder performance and

enabling self-rejuvenating asphalt systems [5].

Despite the progress achieved through experimental approaches, conventional pavement engineering practice remains largely dependent on deterministic laboratory testing. While such methods provide reliable results for specific material samples, they often fail to capture the inherent spatial and temporal variability associated with pavement materials [6]. This variability arises from multiple sources, including heterogeneity in material composition, construction conditions, and environmental influences. As a consequence, standardized testing procedures that rely on averaged measurements may overlook localized weaknesses or critical zones within pavement structures [7]. The issue becomes particularly pronounced in the case of recycled materials such as waste frying oil, where the chemical composition can vary significantly depending on the source of the original oil, frying conditions, and repeated heating cycles, leading to considerable uncertainty in performance prediction [8]. In parallel, sustainability-driven developments in asphalt technology, including warm mix asphalt production, have further emphasized the need for more flexible and efficient evaluation approaches. For instance, Gokalp and Yani [9] investigated pine gum wax modification as a means of reducing production temperatures and improving environmental performance. Within the broader context of Industry 4.0 and intelligent engineering systems, these challenges have highlighted the limitations of purely empirical “trial-and-error” methodologies and underscored the need for modelling strategies capable of systematically addressing material variability [10].

Hybrid modelling approaches provide a system-oriented pathway for addressing these limitations by enabling the structured integration of physics-based constraints with data-driven stochastic representations. By incorporating experimentally derived boundary conditions into computational frameworks, methods such as Monte Carlo simulation allow for the exploration of a wide range of potential material states without the need for extensive laboratory testing. At the same time, the complex interdependencies among key rheological properties—such as the inverse relationship between penetration and softening point—necessitate the use of advanced statistical tools capable of preserving physically meaningful correlations. Copula-based modelling offers a robust mechanism for capturing such nonlinear dependency structures while maintaining consistency with engineering principles.

In this study, a hybrid modelling architecture is established to predict the rheological performance of asphalt binders modified with waste frying oil. Rather than generating new experimental data, the approach integrates literature-derived boundary conditions with stochastic simulation components, including large-scale Monte Carlo sampling and Gaussian copula-based dependency modelling. This integrated structure enables the construction of statistically convergent and physically consistent synthetic datasets, providing a basis for systematic performance evaluation under uncertainty. The objective is not only to improve predictive capability, but also to demonstrate how hybrid modelling structures can be embedded within engineering-oriented analytical systems. In this context, the proposed architecture offers a structured representation of material behaviour that can support the development of digital twin frameworks for pavement performance assessment.

2 Methodology

The core of this research lies in the transition from a purely empirical approach to a hybrid stochastic framework. This section details the computational steps used to model the performance of waste frying oil-modified binders by integrating physical boundaries with probabilistic simulations to bridge the gap between limited laboratory data and large-scale performance prediction. As computational simulations become more prevalent, there is a growing recognition of the need to rigorously quantify uncertainties associated with model parameters and experimental measurements [11, 12].

2.1 Definition of Material Boundary Conditions

Table 1. Input boundary conditions used in the stochastic simulation

Variable	Symbol	Unit	Range / Value	Source / Basis
Waste frying oil dosage	WFO	% by bitumen weight	0–5%	Based on [2]
Penetration	PEN	dmm	47–113	Experimental benchmark
Softening point	SP	°C	43–54	Experimental benchmark
Simulation size	N	iterations	1,000,000	Monte Carlo design
Waste frying oil clipping range	—	%	0–10%	Physical constraint

To ensure that the simulation remains grounded in engineering reality, initial boundary conditions were established based on established literature and previous experimental benchmarks [2]. The primary input variables include the waste frying oil dosage (ranging from 0% to 5% by weight of bitumen), penetration values (47–113 dmm), and softening point temperatures (43–54 °C). These empirical “anchor points” serve as the physical constraints and the statistical basis for the stochastic generator, addressing the need for experimental methods that accurately capture

initial material behavior before structural optimization under uncertainty [13]. The input boundaries are given in Table 1.

2.2 Copula-Based Correlation Modeling

A critical challenge in synthetic data generation is maintaining the physical interdependency between variables, as conventional correlation metrics like Pearson’s coefficient may not fully capture nonlinear and non-Gaussian relationships in complex materials [14]. In bitumen rheology, penetration and softening points are inversely correlated; an increase in the former typically necessitates a decrease in the latter. To preserve this relationship, a Gaussian copula function was utilized to model the joint distribution of these properties. Unlike standard random sampling, the copula was calibrated using rank correlation derived from empirical measurements. This approach enables more accurate uncertainty quantification by explicitly modeling the dependency structure among multiple physical parameters irrespective of their marginal distributions [15]. This ensures that the generated synthetic dataset reflects the actual physical “softening-stiffening” trade-off of asphalt binders rather than purely random statistical noise.

2.3 Data Generation via Monte Carlo Simulation

Once the copula-based dependency structure was established, Monte Carlo simulation was employed to expand the dataset. A total of 1,000,000 iterations were performed to ensure statistical convergence and to generate probabilistic distributions of outcomes rather than single-point estimates [13]. For each iteration, the modifier dosage was treated as a random variable, and the corresponding bitumen properties were sampled from the copula-defined probability space. This hybrid approach allows for the exploration of “edge cases” and material uncertainties—such as variability in waste oil chemical compositions—that are often missed in limited laboratory trial batches ($n = 4$). To filter stochastic noise and identify the most representative engineering trend, a mode-based master curve was constructed using incremental dosage bins ($\Delta 0.2$). For each bin, the most frequent observation (mode) was extracted and smoothed using a Savitzky-Golay filter to derive the final predictive curves.

2.4 Algorithm Workflow and Implementation

The hybrid framework was implemented through a four-step computational workflow:

- Input phase: Definition of marginal distributions (mean and standard deviation) for the impact of waste frying oil based on literature.
- Correlation phase: Application of the Gaussian copula to lock the dependency between penetration and softening point.
- Simulation phase: Execution of 1,000,000 Monte Carlo simulation iterations within the copula’s constraints to generate physically consistent synthetic data points.
- Optimization phase: Implementation of non-negativity constraints and boundary clipping (0–10% waste frying oil) to ensure the physical validity of the synthetic outcomes.
- Output phase: Generation of the final dataset for performance prediction and validation against original empirical benchmarks.

The Algorithm 1 is given as follows.

Algorithm 1 Hybrid stochastic predictive framework for bio-modified bitumen

Inputs:

- Experimental Benchmarks (X_{emp}): Modifier Dosage (W), Penetration (P), Softening Point (SP).
- Simulation Parameters: Sample size ($N = 10^6$), Bin size ($\Delta = 0.2$), Regularization factor ($\lambda = 0.95$).

Output:

- Stochastic Performance Envelope (E).
- Mode-Based Master Curve Equations (f_{pen}, f_{soft}).

Step 1: Dependency Calibration (Gaussian Copula Integration)

- (a) Compute the Spearman rank correlation matrix (ρ_{raw}) from X_{emp} .
- (b) Apply softening factor (λ) to derive the regularized correlation matrix:

$$\Sigma = I + (\rho_{raw} - I)\lambda$$

- (c) Define the marginal distributions for W , P , and SP using empirical means (μ) and standard deviations (σ).
-

Step 2: Stochastic Expansion (Monte Carlo Execution)

- (a) Generate N samples from a Multivariate Normal Distribution: $Z \sim \mathcal{N}(0, \Sigma)$.
- (b) Apply Probability Integral Transform (Cumulative Distribution Function) to map Z into Uniform space $U \in [0, 1]$.
- (c) Transform U into the physical domain using Inverse Cumulative Distribution Function (Percent Point Function):

$$X_{\text{sim}} = \Phi^{-1}(U, \mu, \sigma)$$

- (d) **Constraint Enforcement:** Clip W to the physical bound $[0, 10\%]$ and enforce non-negativity.

Step 3: Mode-Based Feature Extraction

- (a) Partition X_{sim} into incremental bins of size Δ .
- (b) For each bin i :
 - Identify the statistical Mode (most frequent value) for P and SP .
 - Store as discrete Master Points: $M_i = (W_i, \text{mode}(P)_i, \text{mode}(SP)_i)$.

Step 4: Trend Optimization and Validation

- (a) Apply a 2nd-order Savitzky-Golay Filter to M_i to eliminate high-frequency stochastic noise.
 - (b) Perform a Polynomial Regression fit to derive the continuous Master Curve equations.
 - (c) Compute Accuracy (R^2) by benchmarking predicted values against the original X_{emp} data.
-

3 Results and Computational Analysis

3.1 Stochastic Data Expansion and Probabilistic Envelopes

The hybrid framework successfully transformed four empirical laboratory benchmarks into a high-density synthetic dataset comprising 1,000,000 iterations. As illustrated in Figure 1, the Gaussian copula-based Monte Carlo simulation generated a comprehensive probabilistic envelope (the “gray cloud”) that captures the inherent variability of waste frying oil modification.

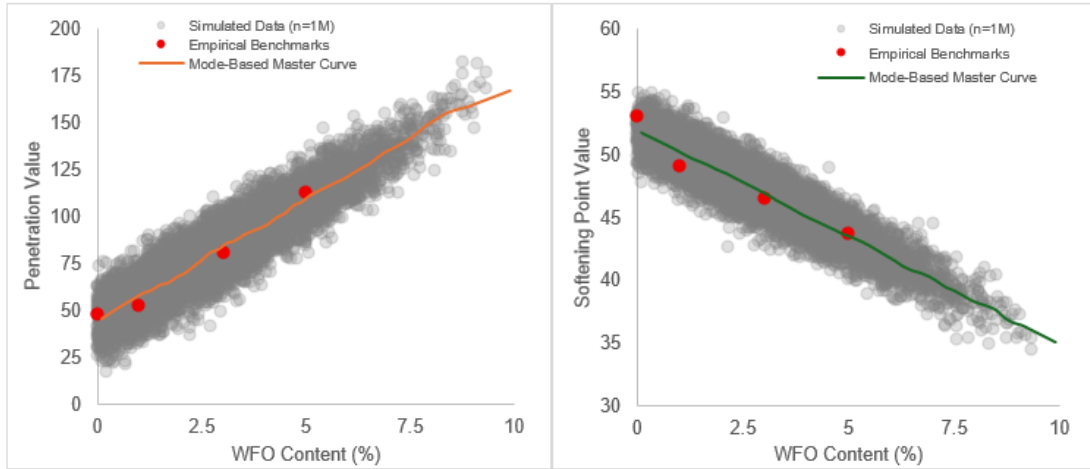


Figure 1. Hybrid stochastic prediction model for waste frying oil-modified bitumen

Unlike traditional deterministic models, this stochastic approach visualizes the confidence intervals of binder performance. The high density of simulated points surrounding the red empirical anchors confirms that the model maintains physical consistency while accounting for the chemical complexity and potential batch-to-batch variability associated with bio-based waste modifiers.

3.2 Monte Carlo Convergence and Seed Robustness

To further evaluate the numerical stability of the stochastic simulation, the Monte Carlo procedure was repeated using ten independent random seeds, including the seed used in the main analysis. For each seed, cumulative mean values of penetration and softening point were calculated at progressively increasing valid sample sizes after physical constraint filtering. As shown in Figure 2, the confidence intervals narrowed substantially as the number of valid Monte Carlo samples increased, indicating that the simulation output stabilized with increasing sample size. This result confirms that the generated stochastic envelope is not dependent on a single random initialization and supports the adequacy of the selected simulation scale.

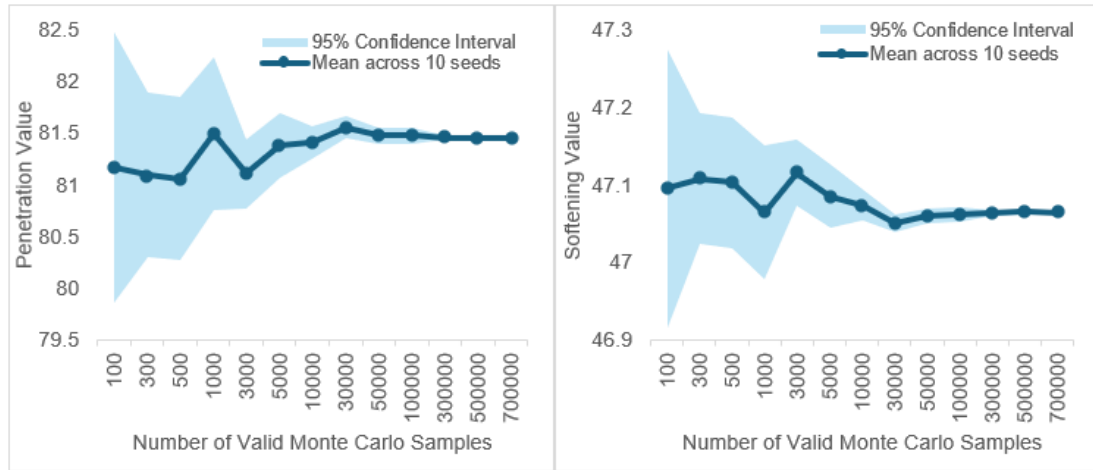


Figure 2. Monte Carlo convergence behavior with 95% confidence intervals (CI) across ten independent random seeds

3.3 Predictive Master Curve Construction

To derive a practical engineering tool from the stochastic data, a mode-based master curve was constructed. By analyzing the most probable outcomes at incremental modifier dosages ($\Delta 0.2$), the model filtered high-frequency statistical noise to reveal the underlying rheological trends. The relationship between the waste frying oil dosage and the binder properties was successfully modeled using second-order polynomial regressions. The resulting master curve equations are defined as follows:

$$y_{\text{penetration}} = -0.0717\text{WFO}^2 + 13.5495\text{WFO} + 43.2675 \quad (1)$$

$$y_{\text{softening}} = -0.0031\text{WFO}^2 - 1.6762\text{WFO} + 51.9254 \quad (2)$$

These equations allow for the rapid estimation of binder consistency across a continuous range (0–10% waste frying oil), effectively serving as a digital twin of the bio-modified bitumen testing process.

3.4 Statistical Validation and Model Accuracy

The reliability of the hybrid framework was validated by comparing the predicted values against the original experimental benchmarks. The model demonstrated exceptional goodness-of-fit, as evidenced by the high coefficients of determination (R^2).

Table 2. Accuracy of the metrics

Performance Metric	Predictive Accuracy (R^2)
Penetration (dmm)	0.9801
Softening Point ($^{\circ}\text{C}$)	0.9375

As seen in Table 2, the 0.9801 accuracy for penetration indicates that the model explains over 98% of the variance observed in the experimental data. While the softening point exhibited a slightly lower R^2 of 0.9375, this remains well within the acceptable limits for bituminous material modeling, where the fatty acid profiles of vegetable oils often introduce subtle non-linearities in thermal response. The integration of the Savitzky-Golay smoothing filter was instrumental in achieving this stability.

3.5 Sensitivity to Bio-Modifier Dosage

The analysis confirms a high sensitivity of bitumen rheology to waste frying oil content. For every 1% increase in waste frying oil, the penetration increases by approximately 13.5 dmm at lower dosages, signaling a significant rejuvenating and softening effect. Conversely, the softening point decreases at a nearly linear rate of 1.68 $^{\circ}\text{C}$ per 1% of waste frying oil. The mathematical alignment of these two distinct properties proves that “intelligent engineering” tools can effectively replace exhaustive laboratory repetitions, providing a robust decision-support system for sustainable and circular pavement design.

4 Conclusion

This study establishes a hybrid modelling architecture for predicting the rheological performance of waste frying oil-modified asphalt binders. By integrating literature-based boundary conditions with Gaussian copula dependency modelling and Monte Carlo simulation components, the proposed approach generates a probabilistic performance envelope from limited empirical benchmark data. This enables both the central trend and the associated uncertainty of waste frying oil-modified binder behaviour to be captured in a consistent manner.

The results indicate that waste frying oil increases penetration while reducing the softening point, confirming its softening and rejuvenating effect on bitumen. This suggests that waste frying oil can function as a bio-based fluxing or rejuvenating agent, particularly in applications where improved workability and reduced production temperatures are required. However, excessive waste frying oil content may adversely affect high-temperature stability, and dosage selection should therefore balance workability improvement with thermal resistance.

The mode-based master curves provide practical predictive equations for estimating penetration and softening point across a continuous range of waste frying oil contents. The modelling architecture achieves high predictive accuracy, with R^2 values of 0.9801 for penetration and 0.9375 for softening point. In addition, convergence analysis across ten independent random seeds confirms that the Monte Carlo outputs are statistically stable and not dependent on a single random initialization.

Overall, the proposed modelling architecture can serve as a preliminary decision-support tool for screening feasible waste frying oil dosage ranges and reducing unnecessary laboratory repetitions. Nevertheless, it is not intended to replace experimental validation. Final binder and mixture design should remain supported by rheological, mechanical, and durability tests, including dynamic shear rheometer, bending beam rheometer, viscosity, rutting, fatigue, and moisture susceptibility analyses.

4.1 Future Work

Future studies should validate the proposed modelling architecture using larger experimental datasets obtained from different waste frying oil sources, base binder grades, aging conditions, and production batches. Such efforts are necessary to improve the generalizability of the model and to account for the inherent variability of waste-derived oils.

Further work should incorporate additional rheological and performance-related parameters, including viscosity, complex modulus, phase angle, rutting resistance, fatigue behaviour, aging indices, and moisture susceptibility. This would enable the modelling structure to extend beyond binder consistency prediction toward a more comprehensive performance-based assessment.

In addition, the framework can be extended from binder-level prediction to mixture-level and pavement-level performance simulations. Comparative studies involving alternative probabilistic approaches and machine learning methods may also provide further insight into modelling efficiency and predictive robustness.

Finally, integrating the proposed modelling architecture within laboratory databases and field performance monitoring systems could support the development of decision-support and digital twin frameworks for sustainable pavement material design.

Author Contributions

Conceptualization, S.T. and E.E.; methodology, E.E.; software, E.E.; validation, S.T. and E.E.; formal analysis, E.E.; investigation, S.T.; resources, E.E.; data curation, E.E.; writing—original draft preparation, E.E.; writing—review and editing, S.T. and E.E.; visualization, E.E. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, the authors used ChatGPT to refine the grammar and linguistic clarity of the text. The authors take full responsibility for the content of the work.

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